Introduction to Probability Theory

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Outline

1. Probability Space
2. Axioms of Probability
3. Stochastic Processes
4. Brownian Motion and Martingales
5. Stochastic Calculus
6. References

Probability theory is ubiquitous

We all hear the word probability (or chance, likelihood) in our day to day life for example

- There is a **high chance** that it will rain tomorrow.
- The party X is **more likely** to form the government.
- A doctor assess the probability of successes of various alternative remedies.
- A bank wants to calculate the probability a loan will be repaid.
- A drug company calculates the probability of harmful side effects of drugs.

Why Probability ?

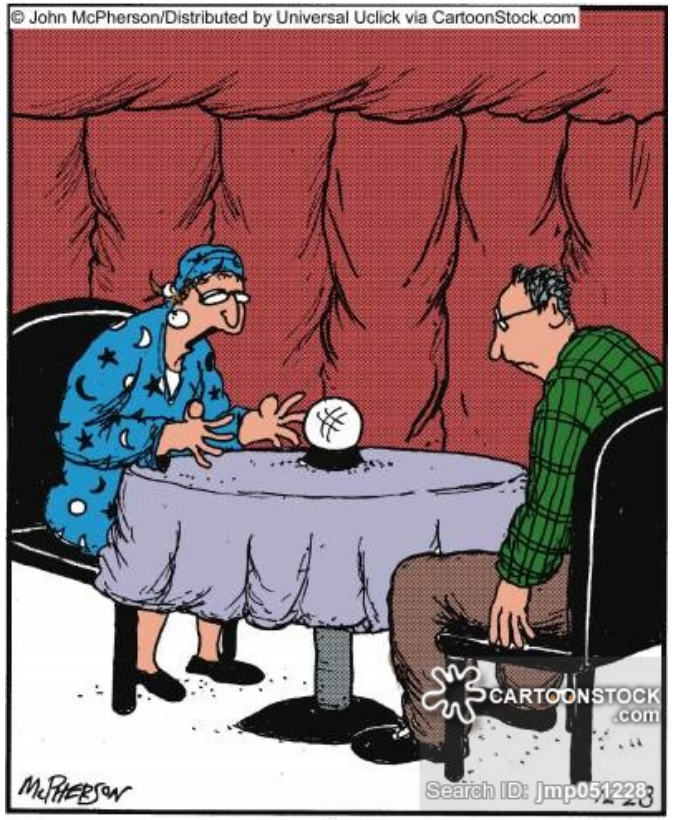
The probable is what usually happens.

Aristotle

... it is a truth very certain that, when it is not in our power to determine the most true opinions we ought to follow the most probable.

Descartes

From time immemorial human beings have been fascinated by the notion of looking into the future - probably because such a capacity would brings them certain advantages.



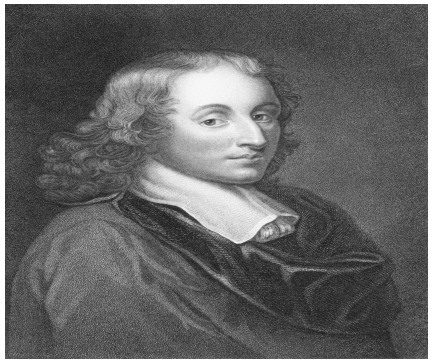
"I see a woman with a very large, round nose ... Oh, wait, that's my reflection. Let me try that again."

Early work on probability

(a) Cardan (1501 - 1576)

(b) Kepler (1571 - 1630)

(a) Galileo (1564 - 1642) and Merton



(b) Pascal (1623 - 1662)

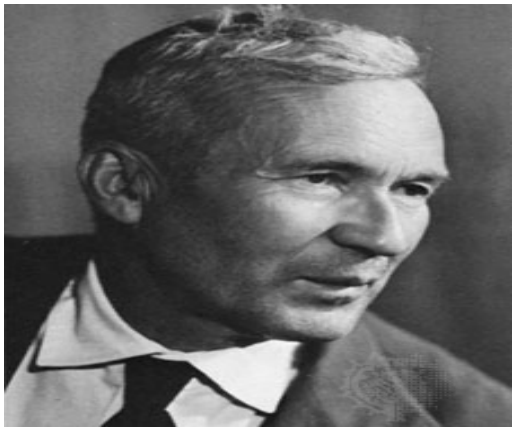
(c) Fermat (1601 - 1665)

de Méré's Problem

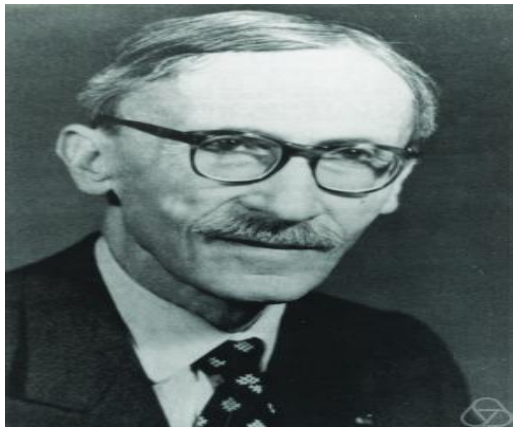
- The probability of getting at least one 'ace' in four roll of a single 6-sided die is higher than the probability of at least one 'double ace' in 24 throw of two dice.
- He posed this question to Pascal who took the help of Fermat to solve this.
- de Méreé came to the answer based on his practical experience and was not sure.

Modern Probability

- In the twentieth century it was Kolmogorov (1903 - 1987) who saw the connection between the ideas of Borel and Lebesgue and probability theory.
- He gave the probability theory its rigorous measure theory basis.
- After the fundamental work of Kolmogorov, Paul Lévy (1886-1971) set the tone for modern probability with his seminal work on Stochastic Processes, Characteristic Function and Limit Theorems.



(a) Kolmogorov (1903 - 1987)



(b) Lévy (1886 - 1971)

Sample Space

In a random experiment a sample space is a collection of all possible outcomes.

- The sample space is generally denoted by Ω.
- The outcomes must be mutually exclusive.
- The outcomes must be collectively exhaustive.

Example

- In a coin toss $\Omega = \{H, T\}$
- In a die roll $\Omega = \{1, 2, 3, 4, 5, 6\}$
- $\Omega = [0, 2\pi]$ if we are interested in angle of rotation in a wheel.

The Events

- An **event** is a property which can be observed either to hold or not to hold after the experiment is done.
- An event is a subset of Ω.
- For two events A and B, the event "*A or B*" is interpreted as the union $A \cup B$.
- The contrary event is the complement set A^c.
- The event "*A and B*" is interpreted as the set $A \cap B$.
- The impossible event is the empty set ϕ
- An **elementary event** is a "singleton" i.e. $\{\omega\}$.

Axioms of Probability

Definition (Algebra)

Let Ω be a non-empty set, and let $\mathcal{F}$ be a collection of subsets of Ω. We say that $\mathcal{F}$ is a an algebra provided that:

i. $\Omega \in \mathcal{F}$.

ii. whenever a set $A \in \mathcal{F}$, its complement A^c also belongs to $\mathcal{F}$.

iii. if $A_1, A_2, \ldots A_n \in \mathcal{F}$ then $\cup_{i=1}^{n} A_i \in \mathcal{F}$.

Example

- $\mathcal{F} = \{\phi, \Omega\}$
- $\mathcal{F} = \{\phi, \Omega, A, A^c\}$.

Definition (σ-algebra)

Let Ω be a non-empty set, and let $\mathcal{F}$ be a collection of subsets of Ω. We say that $\mathcal{F}$ is a σ-algebra provided that:

i. $\Omega \in \mathcal{F}$.

ii. whenever a set $A \in \mathcal{F}$, its complement A^c also belongs to $\mathcal{F}$.

iii. if $A_1, A_2, \ldots A_n, \ldots \in \mathcal{F}$ then $\cup_{n=1}^{\infty} A_n \in \mathcal{F}$.

Example

- $\mathcal{F} = \{\phi, \Omega\}$
- $\mathcal{F} = \{\phi, \Omega, A, A^c\}$.

Example

Let $\mathcal{F}$ be a σ-algebra on $\Omega = [0,1]$ such that $[\frac{1}{n+1}, \frac{1}{n}] \in \mathcal{F}$ for $n = 1, 2, \ldots$. Show that

a. $\{0\} \in \mathcal{F}$
b. $\{\frac{1}{n} : n = 2, 3, \ldots\} \in \mathcal{F}$.
c. $(\frac{1}{n}, 1] \in \mathcal{F}$ for all n
d. $(0, \frac{1}{n}] \in \mathcal{F}$ for all n.

We have

a. $\{0\} = [0,1] \setminus \cup_{n=1}^{\infty} [\frac{1}{n+1}, \frac{1}{n}] \in \mathcal{F}$.
b. $\{\frac{1}{n} : n = 2, 3, \ldots\} = \cup_{n=2}^{\infty} \left([\frac{1}{n+1}, \frac{1}{n}] \cap [\frac{1}{n}, \frac{1}{n-1}] \right) \in \mathcal{F}$.
c. $(\frac{1}{n}, 1] = \cup_{i=1}^{n-1} [\frac{1}{i+1}, \frac{1}{i}] \setminus [\frac{1}{n+1}, \frac{1}{n}] \in \mathcal{F}$.
d. $(0, \frac{1}{n}] = \cup_{i=n}^{\infty} [\frac{1}{i+1}, \frac{1}{i}] \in \mathcal{F}$.

Connection between algebra and σ-algebra

- Any σ-algebra is an algebra which is evident by taking $A_{n+1} = A_{n+2} = \cdots = \phi$.
- An algebra needs not to be a σ-algebra

Example

Let Ω be the set of all natural numbers. Let $\mathcal{F}$ be the family of all subsets which are either finite or have a finite complement. Then $\mathcal{F}$ is an algebra but not a sigma algebra.

Proof.

Note that $\phi \in \mathcal{F}$. If $A \in \mathcal{F}$, then A is finite or A^c is finite. Hence $A^c \in \mathcal{F}$. If $A_1, A_2 \in \mathcal{F}$, then following possibilities are there
A_1 finite, A_2 finite which implies $A_1 \cup A_2$ is finite;
A_1 finite, A_2^c finite which implies $(A_1 \cup A_2)^c = A_1^c \cap A_2^c \subseteq A_2^c$ is finite;
A_1^c finite, A_2 finite which implies $(A_1 \cup A_2)^c = A_1^c \cap A_2^c \subseteq A_1^c$ is finite;
A_1^c finite, A_2^c finite which implies $(A_1 \cup A_2)^c = A_1^c \cap A_2^c$ is finite.
Which in general implies $\mathcal{F}$ is an algebra. $\square$

An algebra needs not to be a σ-algebra

- Again let $\Omega = \mathbb{N}$, i.e. the set of natural numbers.
- Let $\mathcal{F}$ be the family of all subsets which are either finite or have a finite complement.
- Let us take $A_n = \{2, 4, 6, \cdots, 2n\} \in \mathcal{F}$.
- Then the countable union $B := \cup_{n=1}^{\infty} A_n = \{2, 4, 6, \cdots\}$ which is an infinite set.
- Further $B^c = \{1, 3, 5, \cdots\}$ is also an infinite set.
- Thus B and $B^c \notin \mathcal{F}$. However, $A_1, A_2, \cdots, A_n, \cdots \in \mathcal{F}$.
- Hence $\mathcal{F}$ is not a σ-algebra.

Definition (Generated σ-algebra)

Let Ω be a set and $\mathcal{B}$ a non-empty collection of subsets of Ω. The smallest σ-algebra containing all the sets of $\mathcal{B}$ is denoted by $\sigma(\mathcal{B})$ and is called the sigma-algebra generated by the collection $\mathcal{B}$.

Example

- $\sigma(A) = \{\phi, \Omega, A, A^c\}$.
- $\sigma(A, B) = \{\phi, \Omega, A, A^c, B, B^c, A \cap B, A \cup B, A \cap B^c, A \cup B^c, A^c \cap B, A^c \cup B\}$.
- It always exists because 2^Ω is a σ-algebra and the intersection of a family of σ-algebras is again a σ-algebra.

Definition (Borel Sets)

The sets which are obtained from countable unions, intersections, and complements of open sets are called the “Borel Sets”.

Definition (Borel σ-algebra)

Borel σ-algebra on the real line, is the smallest σ-algebra containing all the open sets, or equivalently containing all the closed sets.

Different way to generate Borel σ-algebra

- using open sets.
- using closed sets.
- using open intervals.
- using closed intervals.
- using sets of the form $(-\infty, a],\ a \in Q$.
- using the half open intervals i.e. $(a, b]$ which follows using $(a, b) = \cup_{n=1}^{\infty}(a, b - 1/n]$.

Probability Space

Definition (Probability Space)

Let Ω be a non-empty set, and let $\mathcal{F}$ be a σ-algebra of subsets of Ω. Probability is a function, that, to every subset $A \in \mathcal{F}$, assigns a number in $[0, 1]$ called the probability of A and written $\mathbb{P}(A)$. We require

i. $\mathbb{P}(\Omega) = 1$
ii. If $A_1, A_2, \ldots A_n, \cdots \in \mathcal{F}$ are disjoint events, we have

$$\mathbb{P}(\cup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mathbb{P}(A_n),$$

which is called **countable additivity**.

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a probability space.

Remark

The property

$$\mathbb{P}(\cup_{i=1}^{n} A_i) = \sum_{i=1}^{n} \mathbb{P}(A_i),$$

is called finite additivity. Every probability measure is finite additive.

Some properties of probability measures

It is easy to prove the following

- $\mathbb{P}(\phi) = 0$.
- $\mathbb{P}(A) \leq 1$.
- If $A \subset B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.
- $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$ (inclusion-exclusion principle).
- $\mathbb{P}(A \cap B^c) = 1 - \mathbb{P}(A \cap B)$.
- $\mathbb{P}(\cup_{n=1}^{\infty}) \leq \sum_{n=1}^{\infty} \mathbb{P}(A_n)$.

Proposition

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Then if $A_n \in \mathcal{F}$ and $A_n \uparrow A$, then $\mathbb{P}(A_n) \uparrow \mathbb{P}(A)$.

Proof.

The notation $A_n \uparrow A$, means that $A_n \subset A_{n+1}$ and $\cup_{n=1}^{\infty} A_n = A$. Define

$$\begin{aligned} B_1 &= A_1, \\ B_2 &= A_2 \backslash A_1 \\ &\vdots \\ B_n &= A_n \backslash A_{n-1}. \end{aligned}$$

Thus $\cup_{n=1}^{\infty} B_n = A$ and the events are pairwise disjoint. Using countable additivity, we have

$$\mathbb{P}(A) = \lim_{n \to \infty} \sum_{j=1}^{n} \mathbb{P}(B_j).$$

Also, $\sum_{j=1}^{n} \mathbb{P}(B_j) = \mathbb{P}(A_n)$, hence we deduce that $\lim_{n \to \infty} \mathbb{P}(A_n) = \mathbb{P}(A)$. $\square$

Defining probabilities on a finite or countable space

For a finite or countable sample space Ω, we can take the σ-algebra = $\mathcal{F} = 2^{\Omega}$. Note that 2^{Ω} is always a σ-algebra.

- We have $\phi \in 2^{\Omega}$.
- If $A \in 2^{\Omega}$, then $A \subseteq \Omega$ and hence $A^c \subseteq \Omega$. Hence $A^c \in 2^{\Omega}$.
- If $A_n, n \geq 1$ be a sequence of subset of Ω, then $\cup_{n=1}^{\infty} A_n \subseteq \Omega$ and hence also in 2^{Ω}.

The probability measure $\mathbb{P}$ on Ω is characterized by its values on the singletons $\omega \in \Omega$ i.e. by using $p_{\omega} = \mathbb{P}(\{\omega\}),\ \omega \in \Omega$.

Random Variable

Definition (Random Variable)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A random variable is a real-valued function X defined on Ω such that for every Borel subset B of $\mathbb{R}$ the subset $\{\omega : X(\omega) \in B\} \in \mathcal{F}$.

Example

Let $\Omega = \{H, T\}$ and $\mathcal{F} = \{\phi, \Omega, H, T\}$. Suppose $X(H) = 1$ and $X(T) = 1$. Then for some Borel set B

$$X^{-1}(B) = \{\omega : X(\omega) \in B\} = \begin{cases} \{H\} & \text{if } 1 \in B \\ \{T\} & \text{if } 0 \in B \\ \phi & \text{if } 0, 1 \notin B \\ \Omega & \text{if } 0, 1 \in B \end{cases}$$

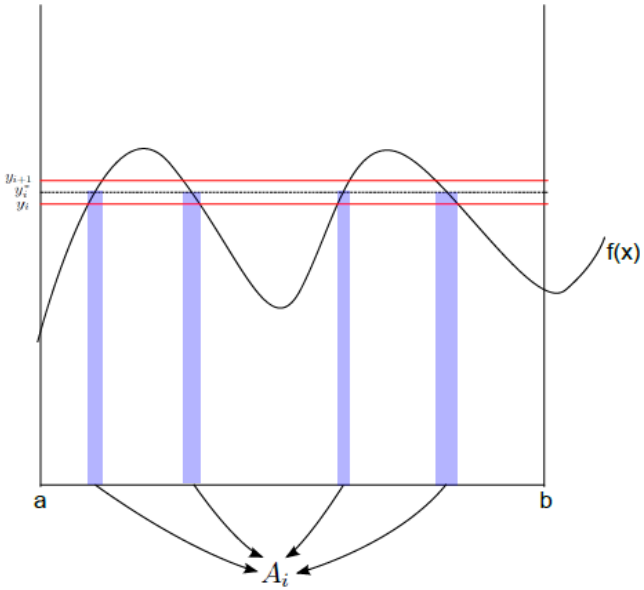
Random variable example

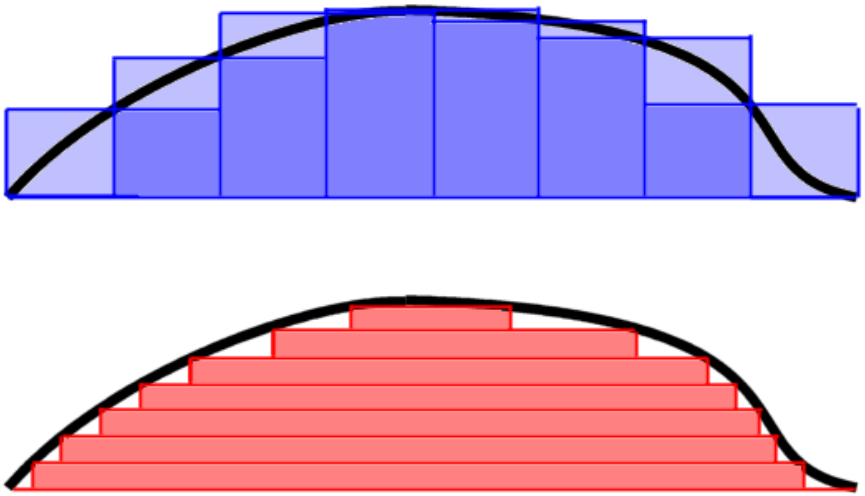
Example

Let $\Omega = \{1, 2, 3, 4\}$ and $\mathcal{F} = \{\phi, \Omega, \{1\}, \{2, 3, 4\}\}$. Is $X(\omega) = 1 + \omega$ is a random variable with respect to the σ-algebra $\mathcal{F}$?

Here X is not a random variable since $X^{-1}(\{5\}) = 4 \notin \mathcal{F}$.

Lebesgue Integral





Definition (Expectation)

For a random variable defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ the expectation is defined by the Lebesgue integral

$$\mathbb{E}X = \int_{\Omega} X d\mathbb{P}.$$

Properties

- $\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$ (Linearity).
- If $X \geq 0$, then $\mathbb{E}(X) \geq 0$ (Positivity).
- If h is a convex function, then $h(\mathbb{E}X) \leq \mathbb{E}(h(X))$ (Jensen Inequality).
- $\mathbb{E}(I_A) = \mathbb{P}(A)$

Pascal's Wager - the first use of expectation in decision making

> Pascal argues that a rational person should live as though God exists and seek to believe in God. If God does actually exist, such a person will have only a finite loss (some pleasures, luxury, etc.), whereas they stand to receive infinite gains (as represented by eternity in Heaven) and avoid infinite losses (eternity in Hell)
>
> *Wikipedia*

Let random variable X be the gain of a person in believing in GOD

$$X = \begin{cases} +\infty & \text{with probability } 0.5 \\ -M & \text{with probability } 0.5 \end{cases}$$

Then $\mathbb{E}X = \infty \times 0.5 - M \times 0.5 = \infty$.

Definition (Conditional expectation given a set)

For any random variable X and any event B such that $\mathbb{P}(B) \neq 0$, the conditional expectation of X given B is defined by

$$\mathbb{E}(X|B) = \frac{\mathbb{E}(I_B X)}{\mathbb{P}(B)}.$$

Definition (Conditional expectation given a discrete random variable Y)

For any random variable X and any discrete random variable Y with different values $y_1, y_2, \cdots$. Then the conditional expectation of $\mathbb{E}(X|Y)$ is defined to be random variable such that

$$\mathbb{E}(X|Y)(\omega) = \mathbb{E}(X|\{Y = y_i\}) \text{ for every } \omega \in \{Y = y_i\}.$$

σ-Algebra Generated by a Random Variable

Definition (σ-Algebra Generated by a Random Variable)

Let X be a random variable defined on a non-empty sample space Ω. The σ-algebra generated by X, denoted $\sigma(X)$, is the collection of all subsets of Ω of the form $\{X \in B\}$, where B ranges over the Borel subsets of $\mathbb{R}$.

Example

Consider the random variable $X = 1_{[0,\frac{1}{2}]}$ over $\Omega = [0, 1]$. In this case the σ-algebra generated by X is

$$\sigma(X) = \{\phi, [0, 1], [0, 1/2], (1/2, 1]\}.$$

Example

Let $X : \Omega \to \mathbb{R}$ denoted the number of heads in a coin tossing experiment. Then the sigma algebra generated by X is

$$\sigma(X) = \{\phi, \Omega, \{H\}, \{T\}\}.$$

σ-algebra generated by a set of random variables

Let $X_i : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R})),\ i \in I$ be a collection of random variables. Then the sigma algebra generated by this set is

$$\sigma(X_i, i \in I) = \sigma\left(\cup_{i \in I} \sigma(X_i)\right).$$

Example

Let X_1 and X_2 denote the number of heads in two coin tosses. Then $\Omega = \{HH, HT, TT, TH\}$

$$\begin{aligned}
\sigma(X_i) &= \{\phi, \Omega, \{\omega : X_i(\omega) = 0\}, \{\omega : X_i(\omega) = 1\}\} \\
\sigma(X_1) &= \{\phi, \Omega, \{TT, TH\}, \{HH, HT\}\} \\
\sigma(X_2) &= \{\phi, \Omega, \{HT, TT\}, \{HH, TH\}\}.
\end{aligned}$$

It follows that

$$\sigma(X_1, X_2) = \sigma\left(\{\phi, \Omega, \{TT, TH\}, \{HH, HT\}, \{HT, TT\}, \{HH, TH\}\}\right).$$

Definition (Stochastic Process)

A stochastic process $X(t)$ is a collection of random variables

$$\{X(t), t \in T\} = \{X(t, \omega), t \in T, \omega \in \Omega\}$$

defined on some probability space $(\Omega, \mathcal{F}, P)$.

- A stochastic process is a function of two variables.
- For fixed instant $t = t_0$, it is random variable $X(t_0, \omega), \omega \in \Omega$.
- For fixed $\omega = \omega_0$ it is a function of time such that $X(t, \omega_0), t \in T$.
- For different ω, we get different functions and these functions are called realizations, trajectories or sample paths of the process.

Example

Brownian motion, Poisson process, fractional Brownian motion, Gaussian process, Markov processes, Lévy processes

Definition (BM)

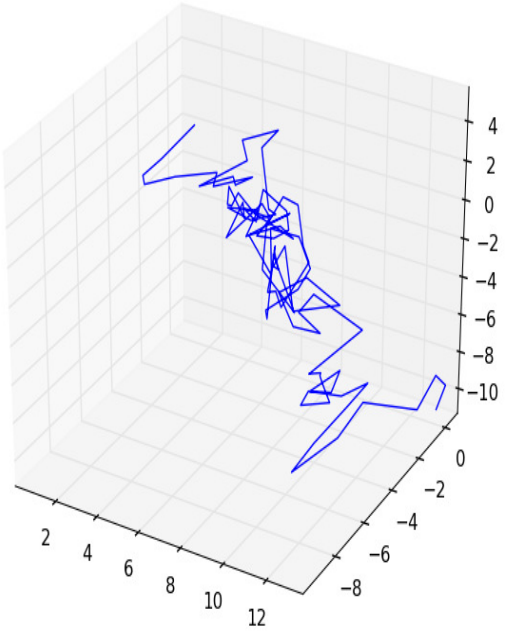
A stochastic process $B(t)$ is said to be a standard BM, if

- $B(0) = 0$ a.s.
- $B(t)$ has independent and stationary increments.
- $B(t) - B(s) \sim \mathrm{N}(0, t - s)$, $0 \leq s < t$.
- Its sample paths are continuous in probability.

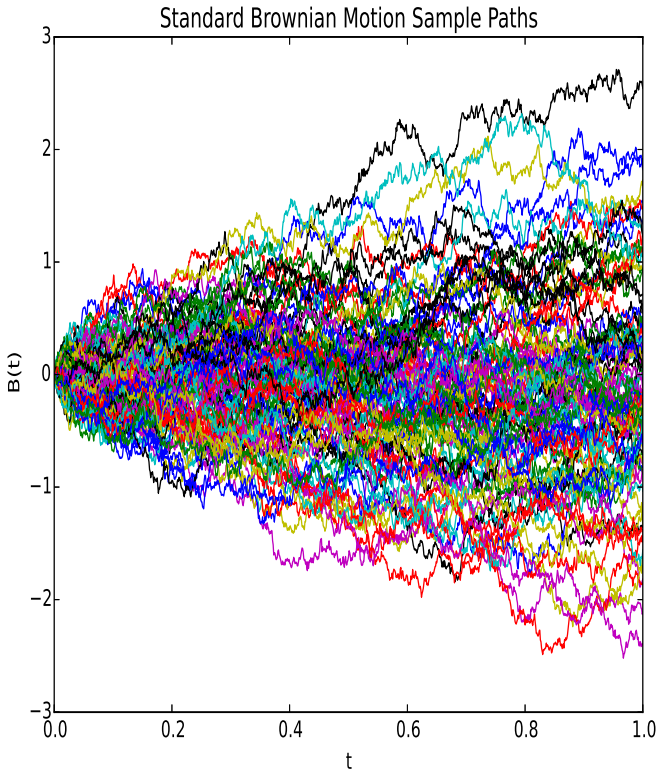
Properties

- $B(t) \sim \mathrm{N}(0, t)$ or $\mathbb{E}B(t) = 0$ and $\mathrm{Var}(B(t)) = t$.
- Further, $\mathrm{Cov}(B(s), B(t)) = \min(s, t)$.
- Brownian motion is a Martingale.

Brownian Motion Sample Trajectories



(a) 3D Brownian Motion

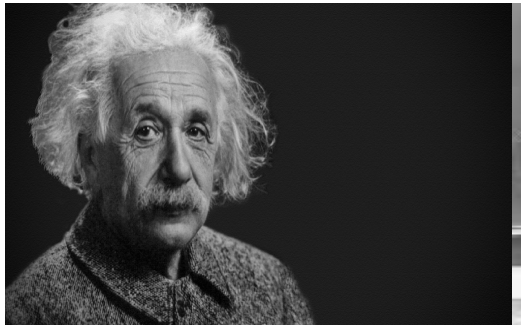


(b) 1D Brownian Motion

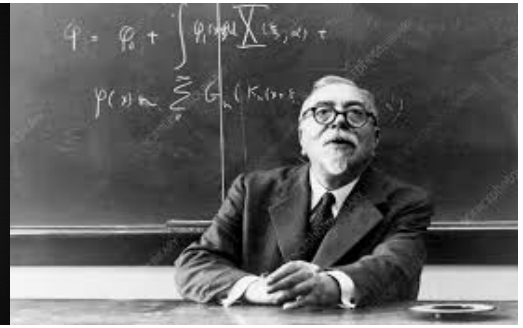
Figure: Brownian Motion Sample Paths

(a) Robert Brown (1827)

Figure: 1



(a) Albert Einstein (1905)



(b) Norbert Wiener (1926)

Figure: 2

Distribution of Brownian Motion

- Because the increments $B(t_1) - B(t_0)$, $B(t_2) - B(t_1)$, $\ldots$, $B(t_m) - B(t_{m-1})$ are independent and normally distributed, the random variables $B(t_1), B(t_2), \ldots, B(t_m)$ are jointly normally distributed.
- $Cov(B(s), B(t) = \min(s, t)$.
- The covariance matrix for the vector $(B(t_1), B(t_2), \ldots, B(t_m))$ is

$$C = \begin{pmatrix} t_1 & t_1 & \cdots & t_1 \\ t_1 & t_2 & \cdots & t_2 \\ \vdots & \vdots & \ddots & \vdots \\ t_1 & t_2 & \cdots & t_m \end{pmatrix}$$

Definition (Filtration)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A filtration on $(\Omega, \mathcal{F}, \mathbb{P})$ is an increasing family $(\mathcal{F}_t), t \geq 0$ of sub-σ-algebras of $\mathcal{F}$. In other words, for each t, $\mathcal{F}_t$ is a σ-algebra included in $\mathcal{F}$ and if $s \leq t$, $\mathcal{F}_s \subseteq \mathcal{F}_t$. A probability space $(\Omega, \mathcal{F}, \mathbb{P})$ endowed with a filtration $(\mathcal{F}_t), t \geq 0$ is called a filtered probability space.

Filtration Interpretation

A filtration $\mathcal{F}_t$ contains any information that could be possibly asked and answered for the considered random process at step t.

Definition (Natural Filtration)

For a stochastic process, the collection of sub-σ-algebras $\mathcal{F}_t^X = \sigma(X_s,\ s \leq t),\ t \geq 0$ is called a natural filtration.

Definition (Adapted Stochastic Process)

A stochastic process X_t on $(\Omega, \mathcal{F}, \mathbb{P})$ is adapted to the filtration $(\mathcal{F}_t)$ if, for each $t \geq 0$, X_t is $\mathcal{F}_t$-measurable. Note that every stochastic process is adapted to its natural filteration.

Martingale

Definition (Discrete-Time Martingale)

A discrete-time martingale is a discrete-time stochastic process (i.e., a sequence of random variables) $X_1, X_2, X_3, \ldots$, that satisfies for any time n,

$$\mathbb{E}(|X_n|) < \infty$$

$$\mathbb{E}(X_{n+1} \mid X_1, \ldots, X_n) = X_n.$$

Alternatively,

$$\mathbb{E}(X_{n+1} \mid \mathcal{F}_n) = X_n,$$

where $\mathcal{F}_n = \sigma(X_1, X_2, \ldots, X_n)$ is the σ-algebra generated by $X_1, X_2, \ldots, X_n$.

Example

A gambler is playing a fair game, where if he wins, he receives Rs 100 and if he losses, he pays Rs 100. Let S_n denotes his holding after game n, then S_n is a martingale.
Solution: Note that

$$\begin{aligned}\mathbb{E}(S_n \mid X_1, X_2, \ldots, X_{n-1}) &= \mathbb{E}(S_{n-1} + X_n \mid X_1, X_2, \ldots, X_{n-1}) \\ &= S_{n-1} + \mathbb{E}(X_n) = S_{n-1} + \frac{1}{2} \times 100 + \frac{1}{2} \times (-100) = S_{n-1},\end{aligned}$$

hence martingale.

Definition (Martingale)

A continuous time stochastic process $X_t,\ t \geq 0$ is called a martingale with respect to the filtration $\mathcal{F}_t$, if

i. $\mathbb{E}|X_t| < \infty$
ii. $\mathbb{E}(X_t \mid \mathcal{F}_s) = X_s,\ s < t.$

Properties

- Martingale represents a fair game.
- $\mathbb{E}(X_t) = \mathbb{E}(X_0)$.

Definition (Submartingale and Supermartingale)

A continuous time stochastic process $X_t,\ t \geq 0$ is called a submartingale or supermartingale with respect to the filtration $\mathcal{F}_t$, if

i. $\mathbb{E}|X_t| < \infty$ and
ii. $\mathbb{E}(X_t \mid \mathcal{F}_s) \geq X_s,\ s < t$ or $\mathbb{E}(X_t \mid \mathcal{F}_s) \leq X_s,\ s < t$, respectively.

Properties

- Submartingale and supermartingale represent unfair games.
- $\mathbb{E}(X_t) \geq \mathbb{E}(X_0)$ for submartingale and $\mathbb{E}(X_t) \leq \mathbb{E}(X_0)$ for supermartingale.

Proposition

The Brownian motion is a martingale with respect to the natural filtration $\mathcal{F}_t^B$.

Proof.

Note that $\mathbb{E}|B(t)| < \infty$.

$$\begin{aligned}\mathbb{E}(B(t) \mid \mathcal{F}_s^B) &= \mathbb{E}(B(t) - B(s) \mid \mathcal{F}_s^B) + \mathbb{E}(B(s) \mid \mathcal{F}_s^B) \\ &= \mathbb{E}(B(t) - B(s)) + B(s) \\ &= B(s),\end{aligned}$$

hence martingale. $\square$

Proposition

The process $M(t) = B(t)^2 - t,\ t \geq 0$ is a martingale with respect to the filtration $\mathcal{F}_t^B$.

Proof.

$$\begin{aligned}
\mathbb{E}(B(t)^2 - t - (B(s)^2 - s) \mid \mathcal{F}_s^B) &= \mathbb{E}((B(t) - B(s))^2 + 2B(t)B(s) - 2B(s)^2 - t + s \mid \mathcal{F}_s^B) \\
&= \mathbb{E}(B(t) - B(s))^2 + 2B(s)^2 - 2B(s)^2 - t + s \\
&= t - s - t + s = 0,
\end{aligned}$$

hence $\mathbb{E}(B(t)^2 - t \mid \mathcal{F}_s^B) = B(s)^2 - s$. $\square$

Proposition

The process $e^{\sigma B(t)-\frac{1}{2}\sigma^2 t}$, $t \geq 0$ is a martingale with respect to the filtration $\mathcal{F}_t^B$.

Proof.

Note that $B(t) \sim N(0,t)$. For a random variable $X \sim N(\mu, \sigma^2)$, we have

$$\begin{aligned}
\mathbb{E}(e^{uX}) &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ux} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} dx \\
&= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{u(\mu+\sigma z)} e^{-\frac{1}{2}z^2} \sigma dz \quad (\text{put } x = \mu + \sigma z) \\
&= e^{\mu u + \frac{1}{2}u^2\sigma^2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-\sigma u)^2} dz = e^{\mu u + \frac{1}{2}u^2\sigma^2}.
\end{aligned}$$

Thus

$$\begin{aligned}
\mathbb{E}(e^{\sigma(B(t)-B(s))-\frac{1}{2}\sigma^2 t} \mid \mathcal{F}_s^B) &= e^{-\frac{1}{2}\sigma^2 t}\mathbb{E}(e^{\sigma(B(t)-B(s))} \mid \mathcal{F}_s^B) \\
&= e^{-\frac{1}{2}\sigma^2 t}\mathbb{E}(e^{\sigma(B(t)-B(s))}) \\
&= e^{-\frac{1}{2}\sigma^2 t}\mathbb{E}(e^{\sigma B(t-s)}) \\
&= e^{-\frac{1}{2}\sigma^2 t} e^{\frac{1}{2}(t-s)\sigma^2} = e^{-\frac{1}{2}\sigma^2 s},
\end{aligned}$$

hence proved. □

Poisson process is not a martingale

Proposition

The compensated Poisson process $N(t) - \lambda t$ is a martingale, where λ is the intensity parameter. However the standard Poisson process $N(t)$ is not.

Proof.

We have

$$\begin{aligned}\mathbb{E}(N(t) - N(s) - (\lambda t - \lambda s) \mid \mathcal{F}_s^N) &= \mathbb{E}(N(t) - N(s) - (\lambda t - \lambda s)) \\ &= \mathbb{E}N(t) - \mathbb{E}N(s) - \lambda t + \lambda s = 0,\end{aligned}$$

hence $N(t) - \lambda t$ is a martingale. However

$$\mathbb{E}(N(t) - N(s) \mid \mathcal{F}_s^N) = \mathbb{E}(N(t) - N(s)) = \lambda(t - s) \neq 0,$$

hence $N(t)$ is not a martingale. However $N(t)$ is a submartingale. $\square$

Definition (BM with drift)

Brownian motion with drift $B_\mu(t)$ is defined by

$$B_\mu(t) = \mu t + \sigma B(t), \ \mu \in \mathbb{R}.$$

Properties

- $B_\mu(t) \sim \mathrm{N}(\mu t, \sigma^2 t)$ or $\mathbb{E}B(t) = \mu t$ and $\mathrm{Var}(B(t)) = \sigma^2 t$.
- Further, $\mathrm{Cov}(B(s), B(t)) = \sigma^2 \min(s, t)$.
- Brownian motion with drift is not a Martingale.
- Brownian motion with positive drift is a submartingale.
- Brownian motion with negative drift is a supermartingale.

Steps for simulating Brownian motion

The Brownian motion with drift can be described by the following SDE $dB_\mu(t) = \mu dt + \sigma dB(t)$.

- The discretized version is $\Delta B_\mu(t) = \mu \Delta t + \sigma \Delta B_\mu(t)$.
- $\Delta B_\mu(t) \sim N(\mu \Delta t, \sigma^2 \Delta t)$, ΔB_μ are independent rvs.
- Thus, $B_\mu(T) = \sum_{t=0}^{T} \Delta B_\mu(t)$.

Simulation Steps

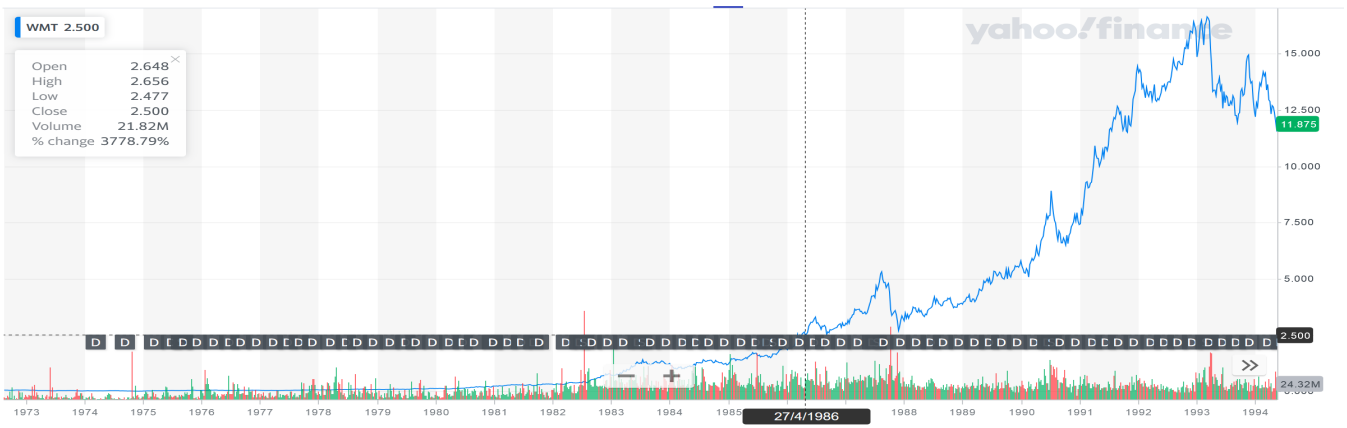
- Discretize the interval $[0, T]$ into uniform intervals of length Δt each.
- Generate $N = T/\Delta t$ normal random numbers with mean $\mu \Delta t$ and variance $\sigma^2 \Delta t$.
- Take the cumulative sum of these random numbers.

Some Stock Prices Charts

(a) Google

(a) Facebook

Some Stock Prices Charts Contd...



(a) Walmart

(a) State Bank of India

Pioneers in Mathematical Finance

(a) Louis Bachelier (1905) (b) Kiyosi Ito (1944)

(a) Black, Scholes (1973) and Merton (b) Harry Markowtiz (1990)

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THANK YOU